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DATABASE:

For the given ODI database perform the following manipulations:
MATCH (match_id, team1, team2, ground, mdate, winner) {primary key → match_id}
PLAYER (p_id, lname, fname, country, yborn, bplace, ftest) {primary key → p_id}
BATTING (match_id, p_id, mts, order, out_type, fow, nruns, nballs, fours, sixes) {primary key → match_id, p_id}
BOWLING (match_id, p_id, novers, maidens, nruns, nwickets) {primary key → match_id, p_id}

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a) Write appropriate DDL statements to create the above database with integrity constraints.

```
SQL> CREATE TABLE MATCH (
  2       match_id VARCHAR2(10) PRIMARY KEY,
  3       team1    VARCHAR2(15),
  4       team2    VARCHAR2(20),
  5       ground   VARCHAR2(20),
  6       mdate    DATE,
  7       winner   VARCHAR2(20)
  8  );
```

```
SQL> CREATE TABLE PLAYER (
  2       p_id    INT PRIMARY KEY,
  3       lname   VARCHAR2(20),
  4       fname   VARCHAR2(20),
  5       country VARCHAR2(20),
  6       yborn   INT,
  7       bplace  VARCHAR2(25),
  8       ftest   INT
  9  );
```

```
CREATE TABLE BATTING (
    match_id VARCHAR2(10),
    p_id     INT,
    mts      INT,
    "order"  INT,
    out_type VARCHAR2(10),
    fow      INT,
    nruns    INT,
    nballs   INT,
    fours    INT,
    sixes    INT,

    PRIMARY KEY (match_id, p_id),
    FOREIGN KEY (match_id) REFERENCES MATCH(match_id),
    FOREIGN KEY (p_id) REFERENCES PLAYER(p_id)
);
```

```
CREATE TABLE BOWLING (
    match_id VARCHAR2(10),
    p_id     INT,
    novers   INT,
    maidens  INT,
    nruns    INT,
    nwickets INT,

    PRIMARY KEY (match_id, p_id),
    FOREIGN KEY (match_id) REFERENCES MATCH(match_id),
    FOREIGN KEY (p_id) REFERENCES PLAYER(p_id)
);
```

b) Create a table COUNTRY with c_id, c_name.

```
SQL> Create Table COUNTRY (c_id int,c_name varchar(40));

Table created.
```

c) Add c_id as PRIMARY KEY for COUNTRY table.

```
SQL> Alter Table COUNTRY
  2  Add constraint pk_country primary key (c_id);

Table altered.
```

d) Drop the table COUNTRY.

```
SQL> drop table COUNTRY;

Table dropped.
```

e) Write appropriate DML statements to insert values in all the tables under the above database.

INSERT INTO **MATCH** VALUES ('2755','India','Sri Lanka','Colombo',TO_DATE('27-08-2008','DD-MM-YYYY'),'India');

INSERT INTO **PLAYER** VALUES (101, 'Kohli', 'Virat', 'INDIA', 1988, 'Delhi', 269);

INSERT INTO **BATTING** VALUES ('2755',101,1,1,'Caught',50,80,60,8,1);

INSERT INTO **BOWLING** VALUES ('2755',103,10,1,55,2);

f) Add a new column to BATTING table.

```
SQL> Alter table BATTING
  2  Add strike_rate DECIMAL(5,2);

Table altered.
```

g) Delete a column from BATTING table.

```
SQL> Alter table BATTING
  2  drop column strike_rate;

Table altered.
```

h) Change any column name of a table.

```
SQL> Alter table MATCH
  2  rename column mdate to match_date
  3  ;

Table altered.
```

i) Get the match id which was played on 27th August, 2008 at Colombo.

```
SQL> Select MATCH_ID
  2  From MATCH
  3  Where GROUND ='Colombo'
  4  and MDATE = DATE '2008-08-27';

MATCH_ID
----------
2755
```

j) Get all match records as the column 'mdate' will be shown as "match_date".

```
SQL> Select MATCH_ID,TEAM1,TEAM2,GROUND,MDATE AS MATCH_DATE,WINNER
  2  from MATCH;

MATCH_ID   TEAM1           TEAM2                GROUND          MATCH_DAT WINNER
---------- --------------- -------------------- --------------- --------- ----------
2324       Pakistan        India                Peshawar        06-FEB-06 Team1
2327       Pakistan        India                Rawalpindi      11-FEB-06 Team2
2357       India           England              Delhi           28-MAR-06 Team1
2377       West Indies     India                Kingston        18-MAY-06 Team2
```

k) Find the player id, first name and last name of all players.

```
SQL> Select P_ID,FNAME,LNAME
  2  From PLAYER;

      P_ID FNAME                LNAME
---------- -------------------- -------------
      8901 Sachin               Tendulkar
      9001 Brian                Lara
      9501 Ricky                Ponting
      9601 Rahul                Dravid
```

l) Insert a new player Michael Clarke with player id 200311 in PLAYER table.

```
SQL> Insert into PLAYER (P_ID,FNAME,LNAME)
  2  values(200311,'MICHAEL','CLARKE');

1 row created.
```

```
SQL> Select * from MATCH;

MATCH_ID   TEAM1           TEAM2                GROUND          MDATE     WINNER
---------- --------------- -------------------- --------------- --------- ----------
2324       Pakistan        India                Peshawar        06-FEB-06 Team1
2327       Pakistan        India                Rawalpindi      11-FEB-06 Team2
2357       India           England              Delhi           28-MAR-06 Team1
2377       West Indies     India                Kingston        18-MAY-06 Team2
2404a      Sri Lanka       India                Colombo         16-AUG-06 Abandoned
2440       India           Australia            Mohali          29-OCT-06 Team2
2449       South Africa    India                Cape Town       26-NOV-06 Team1
2480       India           West Indies          Nagpur          21-JAN-07 Team1
2493       India           West Indies          Vadodara        31-JAN-07 Team1
2520       India           Sri Lanka            Rajkot          11-FEB-07 Team2
2611       England         India                Southampton     21-AUG-07 Team1

MATCH_ID   TEAM1           TEAM2                GROUND          MDATE     WINNER
---------- --------------- -------------------- --------------- --------- ----------
2632       India           Australia            Mumbai          17-OCT-07 Team1
2643       India           Pakistan             Guwahati        05-NOV-07 Team1
2675       Australia       India                Melbourne       10-FEB-08 Team2
2681       India           Sri Lanka            Adelaide        19-FEB-08 Team1
2688       Australia       India                Sydney          02-MAR-08 Team2
2689       Australia       India                Brisbane        04-MAR-08 Team2
2717       Pakistan        India                Karachi         26-JUN-08 Team2
2750       Sri Lanka       India                Colombo         24-AUG-08 Team2
2755       Sri Lanka       India                Colombo         27-AUG-08 Team2
```

```
SQL> Select * from PLAYER;

      P_ID LNAME                FNAME                COUNTRY                   YBORN BPLACE                        FTEST
---------- -------------------- -------------------- -------------------- ---------- ------------------------ ----------
      8901 Tendulkar            Sachin               India                      1973 Mumbai                        1989
      9001 Lara                 Brian                West Indies                1969 Santa Cruz                    1990
      9501 Ponting              Ricky                Australia                  1974 Launceston                    1995
      9601 Dravid               Rahul                India                      1973 Indore                        1996
      9602 Gibbs                Herschelle           South Africa               1974 Cape Town                     1996
      9201 Warne                Shane                Australia                  1969 Melbourne                     1992
      9502 Pollock              Shaun                South Africa               1973 Port Elizabeth                1995
      9903 Vaughan              Michael              England                    1974 Manchester                    1999
      9203 Ul Haq               Inzamam              Pakistan                   1970 Multan                        1992
      9404 Fleming              Stephen              New Zealand                1973 Christchurch                  1994
      9302 Streak               Heath                Zimbabwe                   1974 Bulawayo                      1993

      P_ID LNAME                FNAME                COUNTRY                   YBORN BPLACE                        FTEST
---------- -------------------- -------------------- -------------------- ---------- ------------------------ ----------
      9002 Kumble               Anil                 India                      1970 Bangalore                     1990
      9303 Kirsten              Gary                 South Africa               1967 Cape Town                     1993
      9503 Kallis               Jacques              South Africa               1975 Cape Town                     1995
      9402 Vaas                 Chaminda             Sri Lanka                  1974 Matumgala                     1994
      9202 Muralitharan         Muthiah              Sri Lanka                  1972 Kandy                         1992
      9704 Vettori              Daniel               New Zealand                1979 Auckland                      1997
      2501 Dhoni                MS                   India                      1981 Ranchi                        2005
      2301 Singh                Yuvraj               India                      1981 Chandigarh                    2003
      9603 Ganguly              Saurav               India                      1972 Calcutta                      1996
      9902 Gilchrist            Adam                 Australia                  1971 Bellingen                     1999
      2401 Symonds              Andrew               Australia                  1975 Birmingham                    2004

      P_ID LNAME                FNAME                COUNTRY                   YBORN BPLACE                        FTEST
---------- -------------------- -------------------- -------------------- ---------- ------------------------ ----------
      9900 Lee                  Brett                Australia                  1976 Wollongong                    1999
      9101 Jayasuriya           Sanath               Sri Lanka                  1969 Matara                        1991
      2101 Sehwag               Virender             India                      1978 Delhi                         2001
      9801 Afridi               Shahid               Pakistan                   1980 Khyber Agency                 1998
      9802 Singh                Harbhajan            India                      1980 Jalandhar                     1998
      2701 Kumar                Praveen              India                      1986 Meerut
      2702 Sharma               Ishant               India                      1988 Delhi                         2007
```

```
SQL> Select * from BOWLING;

MATCH_ID   P_ID           NOVERS    MAIDENS      NRUNS   NWICKETS
---------- ---------- ---------- ---------- ---------- ----------
2689       99001              10          0         58          1
2689       24001               3          0         27          1
2689       23001               3          0         15          0
2755       94002               9          1         40          1
2755       92002              10          0         56          1
2755       91001               4          0         29          0
2755       23001              10          0         53          2
2755       98002              10          0         40          3
2689       98002              10          0         44          1
```

```
SQL> Select * from BATTING;

  MATCH_ID       P_ID        MTS        ORD OUT_TYPE          FOW      NRUNS     NBALLS      FOURS      SIXES
---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      2755       2301         12          3 C                  51          0         12          6          0
      2755       2501        104          5 C                 232         71        104         80          4
      2755       9101         85          1 C                  74         60         85         52          8
      2755       9402         23          7 LBW               157         17         23         29          1
      2755       9202         11         11 NO                  1         11          7          0          0
      2689       8901        176          2 C                 205         91        176        121          7
      2689       2301         27          4 C                 175         38         27         38          2
      2689       2501         52          5 C                 240         36         52         37          2
      2689       9902          3          1 C                   2          2          3          3          0
      2689       9501          9          3 C                   8          1          9          7          0
      2689       2401         81          5 C                 123         42         81         56          2

  MATCH_ID       P_ID        MTS        ORD OUT_TYPE          FOW      NRUNS     NBALLS      FOURS      SIXES
---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ---------- ----------
      2689       9900         20          8 B                 228          7         20         12          0
      2689       2701          7          9 C                 255          7          7          7          1
      2755       2701          7          9 B                 257          2          7          6          0
      2689       9802          7          8 LBW               249          3          7          3          0
      2755       9802          7          8 RO                253          2          7          4          0
```